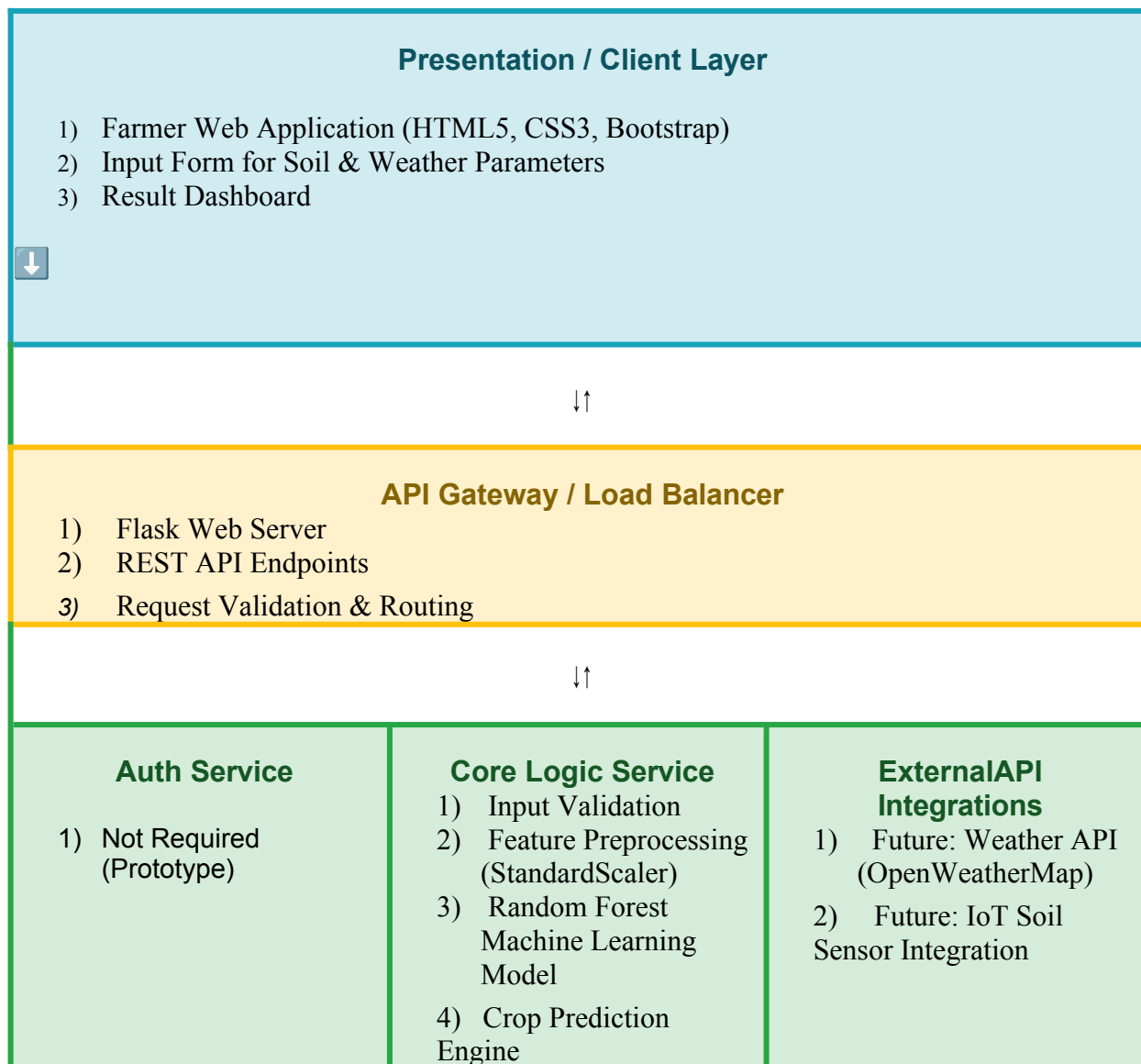


SolutionArchitecture

Date	29 June 2026
TeamID	XXXXXX
ProjectName	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

SolutionArchitectureDiagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



**Data / Storage Layer**

- 1) Crop Recommendation Dataset (CSV)
- 2) Trained Model (.pkl)
- 3) Pandas Data Processing

ComponentDescriptionTable

ComponentName	Description/RoleinArchitecture	TechnologiesUsed
Presentation Layer	Provides a user-friendly web interface where farmers enter soil and weather details and view crop recommendations.	HTML5, CSS3, Bootstrap, JavaScript
API Gateway / Application Layer	Receives user requests, validates inputs, and routes them to the prediction engine.	Flask, Python
Core Logic Service	Processes input data, applies preprocessing, loads the trained Random Forest model, and predicts the best crop.	Python, Scikit-Learn, Pandas, NumPy
External API Integrations	Supports future integration with weather services and IoT-based soil sensors.	OpenWeatherMap API (Future), IoT Sensors (Future)
Data / Storage Layer	Stores the agricultural dataset and trained machine learning model used for prediction.	CSV Dataset, Pickle (.pkl), File Storage